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ODD END SEMESTER EXAMINATION DECEMBER – 2025

Name of the Course: M. Tech Structural Engineering Semester: III
Name of the Paper: Soil-Structure Interaction
Paper Code: MSE315
Time: 3 Hours Maximum Marks: 100

Note:-
(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks

Q1	**(20marks)**	
(a)	Explain the concept of Soil-Structure Interaction (SSI). Why is it important to consider SSI effects in the analysis and design of structures?	
(b)	Define and explain key SSI terminology, including impedance functions, foundation flexibility, kinematic interaction, and inertial interaction. How are these concepts implemented in SSI analysis?	**CO1**
(c)	What are the commonly used "Rule of Thumb" tests or empirical relationships used in SSI evaluation? Discuss key equations related to Average Effective Shear Wave Velocity and their role in SSI modeling.	
Q2	**(20 marks)**	
(a)	Explain the concept of Base Slab Averaging in Soil-Structure Interaction (SSI). Why is it important in determining foundation response, and what are the key equations used to calculate the averaged soil properties beneath the base slab?	
(b)	Discuss the major limitations and issues associated with Base Slab Averaging when used in practical SSI analysis. Provide examples where this method might lead to inaccurate results.	**CO2**
(c)	How is the Depth of Embedment determined for different types of foundations? Explain the criteria and parameters considered in estimating embedment depth with relevant examples.	
Q3	**(20 marks)**	
(a)	Explain the concept of foundation and soil flexibility in soil–structure interaction (SSI). Discuss how foundation flexibility affects the response of a structure under dynamic loading.	**CO3**
(b)	Compare and contrast the following three methods for modeling vertical and rotational foundation flexibility: Method 1: Rigid Foundation on Flexible Soil Method 2: Flexible Foundation on Nonlinear Flexible Soil Method 3: Flexible Foundation on Linear Flexible Soil Include the key assumptions, governing equations, and typical applications of each.	
(c)	A shallow foundation is modeled using Method 1 and Method 3 for vertical and rotational flexibility. Using Method 1 for horizontal flexibility, illustrate how to compute resistance due to passive pressure and friction, with suitable assumptions.	
Q4	**(20 marks)**	
(a)	Explain the concept of period lengthening in soil-structure interaction (SSI). How does soil flexibility affect the fundamental period of a structure compared to a fixed-base condition?	**CO4**

(b)	Using relevant code equations, calculate the fundamental period and design spectral acceleration for a structure on soft soil, given the fixed-base period and soil stiffness values. Compare the results.	
(c)	Differentiate between soil damping and radiation damping in the context of foundation damping. Explain how these mechanisms contribute to the effective damping ratio of a soil-structure system.	
Q5	**(20 marks)**	**CO5**
(a)	Summarize the findings from past analytical comparisons of different basement models. What key differences were observed in predicted responses between simplified and detailed modelling approaches?	
(b)	Discuss the modelling approach for buildings on relatively level grade. Illustrate with an example how basement wall–soil interaction and boundary conditions are represented in such models.	
(c)	Outline the provisions and recommendations on soil–structure interaction (SSI) for basement modelling as per design codes or guidelines. How do these provisions ensure realistic representation of soil confinement and stiffness effects?	